

1. B)

Objective: Find which of the QEC codes can correct for a single spontaneous emission error using the atomic qubit.

QEC codes described in Lec 2:

1. 3-qubit bit-flip repetition code: against X errors
2. 3-qubit phase repetition code: against Z errors
3. 9-qubit Shor's code: protects against X & Z errors.

Now, for a QEC code to be valid, it must satisfy the KL conditions:

$$P E_a^\dagger E_b P = \alpha_{ab} P$$

$$P|\psi_L\rangle = |\psi_L\rangle \quad ; \quad \langle\psi_L|P = \langle\psi_L|$$

$$\underbrace{\langle\psi_L|P} E_a^\dagger E_b \underbrace{P|\psi_L\rangle} = \alpha_{ab} \langle\psi_L|P|\psi_L\rangle$$
$$\langle\psi_L|E_a^\dagger E_b|\psi_L\rangle = \alpha_{ab}$$

$$\underline{\underline{\alpha_{ab}}} = \langle \Psi_L | E_a^\dagger E_b | \Psi_L \rangle$$

this should be the same for $|0_L\rangle$ and $|1_L\rangle$

Let's consider the 3-bit repetition code for bit-flips

$$|\Psi_L\rangle \in \{|0_L\rangle, |1_L\rangle\}$$

$$|0_L\rangle = |000\rangle \quad ; \quad |1_L\rangle = |111\rangle$$

$$K_1, K_2 \in \{E_a\}$$

$$K_1 = |0\rangle\langle 0| + \sqrt{1-p} |1\rangle\langle 1| \quad ; \quad K_2 = \sqrt{p} |0\rangle\langle 1|$$

Let's consider K_1 on qubit 1.

$$E = K_1 \otimes I \otimes I$$

$$\& E^\dagger E = K_1^\dagger K_1 \otimes I \otimes I$$

$$= (|0\rangle\langle 0| + (1-p)|1\rangle\langle 1|) \otimes I \otimes I$$

$$\langle 000 | (|0\rangle\langle 0| + (1-p)|1\rangle\langle 1| \otimes I \otimes I) | 000 \rangle = 1$$

$$\langle 111 | (|0\rangle\langle 0| + (1-p)|1\rangle\langle 1| \otimes I \otimes I) | 111 \rangle = 1-p$$

$$1 \neq 1-p$$

So, KL conditions fail.

Let's consider the 3-bit phase repetition code:

$$|0_L\rangle = \frac{|000\rangle + |011\rangle + |101\rangle + |110\rangle}{2}$$

$$|1_L\rangle = \frac{|100\rangle + |010\rangle + |001\rangle + |111\rangle}{2}$$

actually, let's consider $K_2 = \sqrt{P} |0\rangle \langle 1|$ because it's a simpler operator. $K_2^\dagger K_2 = P |1\rangle \langle 1|$

$$\begin{aligned} & \langle 0_L | K_2^\dagger K_2 \otimes I \otimes I | 0_L \rangle \\ &= \langle 0_L | P | 1 \rangle \langle 1 | \otimes I \otimes I | 0_L \rangle \\ &= P \left(\frac{\langle 000 | + \langle 011 | + \langle 101 | + \langle 110 |}{2} \right) \left(\frac{|101\rangle + |110\rangle}{2} \right) \\ &= \frac{P}{4} \begin{pmatrix} 1 & +1 \end{pmatrix} \quad \text{since the states are orthogonal} \\ &= P/2. \end{aligned}$$

$$\begin{aligned} & \langle 1_L | P | 1 \rangle \langle 1 | \otimes I \otimes I | 1_L \rangle \\ &= P \left(\langle 1_L | \frac{|1100\rangle + |1111\rangle}{2} \right) \\ &= \frac{P}{4} \begin{pmatrix} 0 \end{pmatrix} = P/2. \end{aligned}$$

Because of symmetry K_2 acting on any of the 3 qubits = $p/2$.

Now, let's check the cross-terms: $K_1^\dagger K_2 = \sqrt{p} |0\rangle\langle 1|$

$$\langle 0_L | \sqrt{p} |0\rangle\langle 1| \otimes I \otimes I |1_L\rangle$$

$$\Rightarrow \frac{\sqrt{p}}{2} \langle 0_L | \left(\frac{|000\rangle + |011\rangle}{2} \right)$$

$$\Rightarrow \frac{\sqrt{p}}{2} \neq 0$$

So, KL theorem fails for this as well.

Let's check the Shor's Code:

$$|0_L\rangle = \frac{|000\ 000\ 000\rangle + |000\ 111\ 111\rangle + |111\ 111\ 000\rangle + |111\ 000\ 111\rangle}{2}$$

$$|1_L\rangle = \frac{|111\ 111\ 111\rangle + |111\ 000\ 000\rangle + |000\ 000\ 111\rangle + |000\ 111\ 000\rangle}{2}$$

Let's check for $K_1^\dagger K_1$, $K_1^\dagger K_2$, $K_2^\dagger K_2$, $K_2^\dagger K_1$.

$$K_1^\dagger K_1 = |0\rangle\langle 0| + (1-p)|1\rangle\langle 1|$$

$$\langle 0_L | (|0\rangle\langle 0| + (1-p)|1\rangle\langle 1|) \otimes I \otimes I | 0_L \rangle$$

$$\Rightarrow \langle 0_L | \frac{|0000000000\rangle + |000111111\rangle}{2} + (1-p) \left(\frac{|111111000\rangle + |111000111\rangle}{2} \right) \rangle$$

$$\Rightarrow 1 + (1-p) \frac{2}{2} = 2-p.$$

$$\langle 1_L | (|0\rangle\langle 0| + (1-p)|1\rangle\langle 1|) \otimes I \otimes I | 1_L \rangle$$

$$\Rightarrow \langle 1_L | \frac{1}{2} \text{ the terms in } 1_L + (1-p) \frac{1}{2} \text{ the terms in } 1_L \rangle$$

$$\Rightarrow 1 + (1-p) = 2-p.$$

Let's check for the cross terms:

$$\langle 0_L | (|0\rangle\langle 0| + (1-p)|1\rangle\langle 1|) \otimes I \otimes I | 1_L \rangle$$

$$= \langle 0_L | \frac{1}{2} \text{ the terms in } 1_L + (1-p) \frac{1}{2} \text{ the terms in } 1_L \rangle$$

$$= 1 + (1-p) = 2-p.$$

$$K_1^\dagger K_2 = \sqrt{p} |0\rangle \langle 1|$$

$$\langle 0_L | \sqrt{p} |0\rangle \langle 1| \otimes I \otimes I | 0_L \rangle$$

$$\Rightarrow \langle 0_L | \left(\frac{|011000000\rangle + |011111111\rangle}{2} \right) \sqrt{p}$$

$$\Rightarrow 0$$

$$\langle 1_L | \sqrt{p} |0\rangle \langle 1| \otimes I \otimes I | 1_L \rangle$$

$$\Rightarrow \langle 1_L | \left(\frac{1}{2} \text{ the terms in } 1_L \text{ with 1st bit flipped} \right) \sqrt{p}$$

$$\Rightarrow 0.$$

$$K_2^\dagger K_1 = \sqrt{p} |1\rangle \langle 0|$$

(By symmetry this should also be 0 for both $|0_L\rangle$ and $|1_L\rangle$)

$$K_2^\dagger K_2 = p |1\rangle \langle 1|$$

$$\langle 0_L | p |1\rangle \langle 1| \otimes I \otimes I | 0_L \rangle$$

$$\Rightarrow \langle 0_L | \left(\frac{1}{2} \text{ the terms in } 0_L \right) \cdot p$$

$$\Rightarrow \frac{p}{2}$$

$$\langle 1_L | P | 1 \rangle \langle 1 | I \otimes I | 1_L \rangle$$

$$\Rightarrow \langle 1_L | \left(\frac{1}{2} \text{ the terms in } \right)_{1_L} \cdot P$$

$$\Rightarrow \frac{P}{2}.$$

The cross terms will always be 0.

Shor's Code is a valid QEC code to correct these errors.

Objective: Can we use the pauli measurements to identify & correct these errors?

$$|\psi_L\rangle = \alpha|0_L\rangle + \beta|1_L\rangle$$

$$\begin{aligned} \xrightarrow{\text{acting on the 1st qubit}} K_2^{(1)} |\psi_L\rangle &= \sqrt{P} |0\rangle \langle 1 | (\alpha|0_L\rangle + \beta|1_L\rangle) \\ &= \sqrt{P} \left(\alpha \left| 0_L \text{ with 1st qubit flipped in } \frac{1}{2} \text{ the terms} \right\rangle + \beta \left| 1_L \text{ with 1st qubit flipped in } \frac{1}{2} \text{ the terms} \right\rangle \right) \end{aligned}$$

So this is an X_1 error.

It anticommutes with one of Shor's Code's stabilizers
 $= Z_1 Z_2$

By symmetry a flip on the i^{th} qubit can be identified using $Z_i Z_{i+1}$.

Let's take the K_1 operator:

$$\begin{aligned}
 K_1^{(1)} |\psi_L\rangle &= (|0\rangle\langle 0| + \sqrt{1-p} |1\rangle\langle 1|) |\psi_L\rangle \\
 &= (|0\rangle\langle 0| + \sqrt{1-p} |1\rangle\langle 1|) (\alpha |0_L\rangle + \beta |1_L\rangle) \\
 &= \frac{\alpha}{2} \left(\begin{array}{c} 1/2 \text{ the} \\ \text{terms} \\ \text{in } 0_L \end{array} \right) + \frac{\beta}{2} \left(\begin{array}{c} 1/2 \text{ the} \\ \text{terms} \\ \text{in } 1_L \end{array} \right) \\
 &\quad + \sqrt{1-p} \left[\frac{\alpha}{2} \left(\begin{array}{c} 1/2 \text{ the} \\ \text{terms in} \\ 0_L \end{array} \right) + \frac{\beta}{2} \left(\begin{array}{c} 1/2 \text{ the} \\ \text{terms in} \\ 1_L \end{array} \right) \right]
 \end{aligned}$$

Wait...

the operator $K_1 = \begin{pmatrix} 1 & 0 \\ 0 & \sqrt{1-p} \end{pmatrix}$

$$X = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}; \quad Y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}; \quad I = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

$$Z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}.$$

So, maybe K_1 can be represented in I & Z .

$$K_1 = aI + bZ.$$

$$\begin{pmatrix} a & 0 \\ 0 & a \end{pmatrix} + \begin{pmatrix} b & 0 \\ 0 & -b \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & \sqrt{1-p} \end{pmatrix}$$

$$\Rightarrow a+b = 1 ; \quad a-b = \sqrt{1-p}$$

$$\Rightarrow a = \frac{1+\sqrt{1-p}}{2} ; \quad b = \frac{1-\sqrt{1-p}}{2}$$

$$\text{So; } K_1 = \left(\frac{1+\sqrt{1-p}}{2} \right) I + \left(\frac{1-\sqrt{1-p}}{2} \right) Z$$

So, it's a Z error on a qubit and can be detected using: ZX stabilizers of the Shor's Code.

So, for error correction:

1. Measure the stabilizers / extract the syndrome.
2. Identify which Pauli operators are causing these errors.
3. Correct the error by reapplying the Pauli operator.

Note:

1) Any single-qubit Kraus operator can be expanded in $\{I, X, Y, Z\}$.

- $K_1 = aI + bZ$ (for example)

n-qubit Kraus operators can be represented as a combination of Pauli strings.

2) Kraus operators can't be reversed: $K^{-1} \neq K$.
so the operation can't be undone by applying Kraus again. You have to apply Pauli.

3) Pauli errors can: $X^2 = I$; $Y^2 = I$; $Z^2 = I$.

4) For a case like $K = aI + bZ$, syndrome measurement projects it into one error: I or Z

Then the corresponding pauli is applied to correct it.